Backend Development Progress Report

Project Title

AI Study Material Generator

Team Information

Team Name: CSE4104-7C-T03

Section: 7C

Name	Student ID	Role
Md. Jubayer Hossain	11230121152	Team Leader
Zemima Khan jimu	11230121144	Frontend Developer
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GitHub Repository Link	https://github.com/jubayerhossain5618/AI-Study-Material-Generator

Submission Date: 29/06/2026

1. Backend Technology Stack

The backend of the AI Study Material Generator is developed using the following technologies:

Backend Framework

- Node.js
- Express.js

Database

- MongoDB
- MongoDB Atlas

Authentication

- JWT (JSON Web Token)

Development Tools

- Visual Studio Code
- Postman
- GitHub

2. Backend Project Structure

```
backend/
│
├── config/
│   └── db.js
│
├── controllers/
│   ├── authController.js
│   ├── userController.js
│   ├── documentController.js
│   └── aiController.js
│
├── middleware/
│   ├── authMiddleware.js
│   └── errorMiddleware.js
│
├── models/
│   ├── User.js
│   ├── Document.js
│   ├── Material.js
│   ├── Quiz.js
│   └── ChatHistory.js
│
├── routes/
│   ├── authRoutes.js
│   ├── userRoutes.js
│   ├── documentRoutes.js
│   └── aiRoutes.js
│
├── uploads/
│
├── utils/
│
├── server.js
├── package.json
└── .env
```

*****Screenshot of backend folder structure has given on page 11*****

3. Database Design Summary

The system uses MongoDB as the primary database.

Collections

Users

Field	Type
_id	ObjectId
name	String
email	String
password	String
role	String

Documents

Field	Type
_id	ObjectId
userId	ObjectId
fileName	String
filePath	String
uploadDate	Date

GeneratedMaterials

Field	Type
_id	ObjectId
documentId	ObjectId
materialType	String
generatedContent	String

Quiz

Field	Type
_id	ObjectId
documentId	ObjectId
questions	Array

ChatHistory

Field	Type
_id	ObjectId
userId	ObjectId
message	String
response	String
createdAt	Date

*****Screenshot of mongobd collection has given on page 12-13*****

4. Implemented APIs

Authentication APIs

Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	User login
POST	/api/auth/logout	User logout

User APIs

Method	Endpoint	Description
GET	/api/users/profile	Get user profile
PUT	/api/users/profile	Update user profile

Document APIs

Method	Endpoint	Description
POST	/api/documents/upload	Upload study document
GET	/api/documents	Get uploaded documents
DELETE	/api/documents/:id	Delete document

AI APIs

Method	Endpoint	Description
POST	/api/ai/summary	Generate summary
POST	/api/ai/mcq	Generate MCQs
POST	/api/ai/flashcards	Generate flashcards
POST	/api/ai/quiz	Generate quiz
POST	/api/ai/chat	AI chatbot interaction

*****Screenshot of APIs test has given on page 14-15*****

5. Authentication Workflow

The system uses JWT-based authentication.

Registration Process

1. User enters registration details.
2. Password is encrypted using bcrypt.
3. User information is stored in MongoDB.

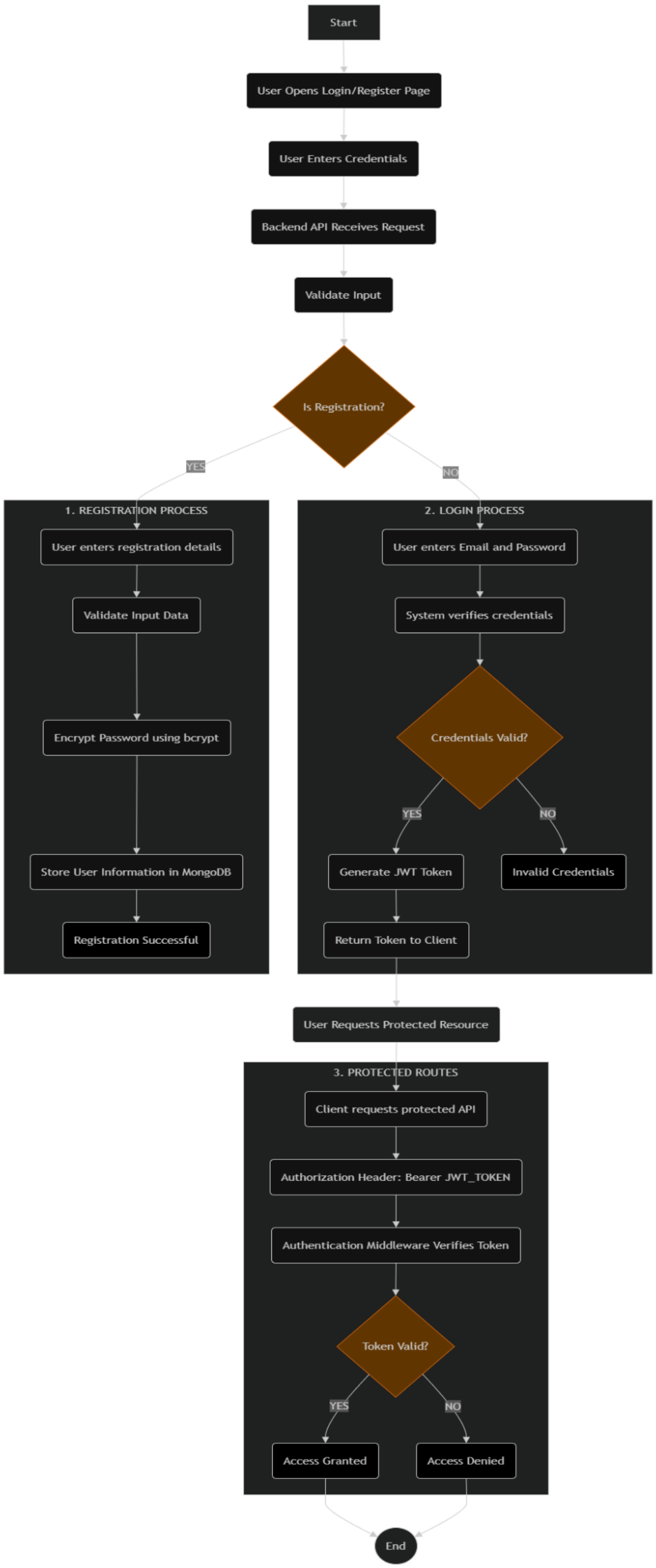
Login Process

1. User enters email and password.
2. Credentials are verified.
3. JWT token is generated.
4. Token is returned to the client.

Protected Routes

Protected APIs require a valid JWT token.

```
Authorization: Bearer <JWT_TOKEN>
```


Start
User Opens Login/Register Page
User Enters Credentials
Backend API Receives Request
Validate Input
Is Registration?
YES
NO
1. REGISTRATION PROCESS
User enters registration details
Validate Input Data
Encrypt Password using bcrypt
Store User Information in MongoDB
Registration Successful
2. LOGIN PROCESS
User enters Email and Password
System verifies credentials
Credentials Valid?
YES
NO
Generate JWT Token
Invalid Credentials
Return Token to Client
User Requests Protected Resource
3. PROTECTED ROUTES
Client requests protected API
Authorization Header: Bearer JWT_TOKEN
Authentication Middleware Verifies Token
Token Valid?
YES
NO
Access Granted
Access Denied
End

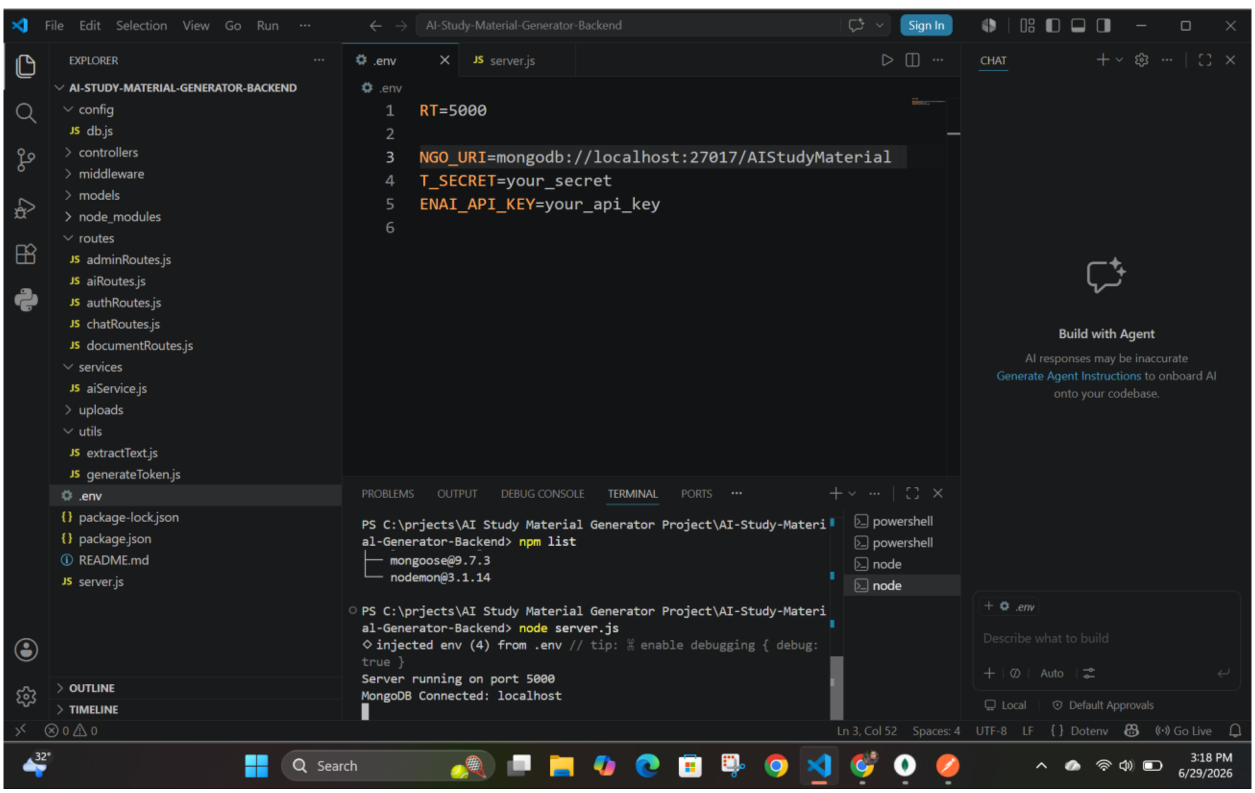
6. Database Connectivity

The backend successfully communicates with MongoDB.

Implemented functionalities:

- Store user information
- Retrieve user information
- Upload documents
- Save generated materials
- Retrieve stored materials

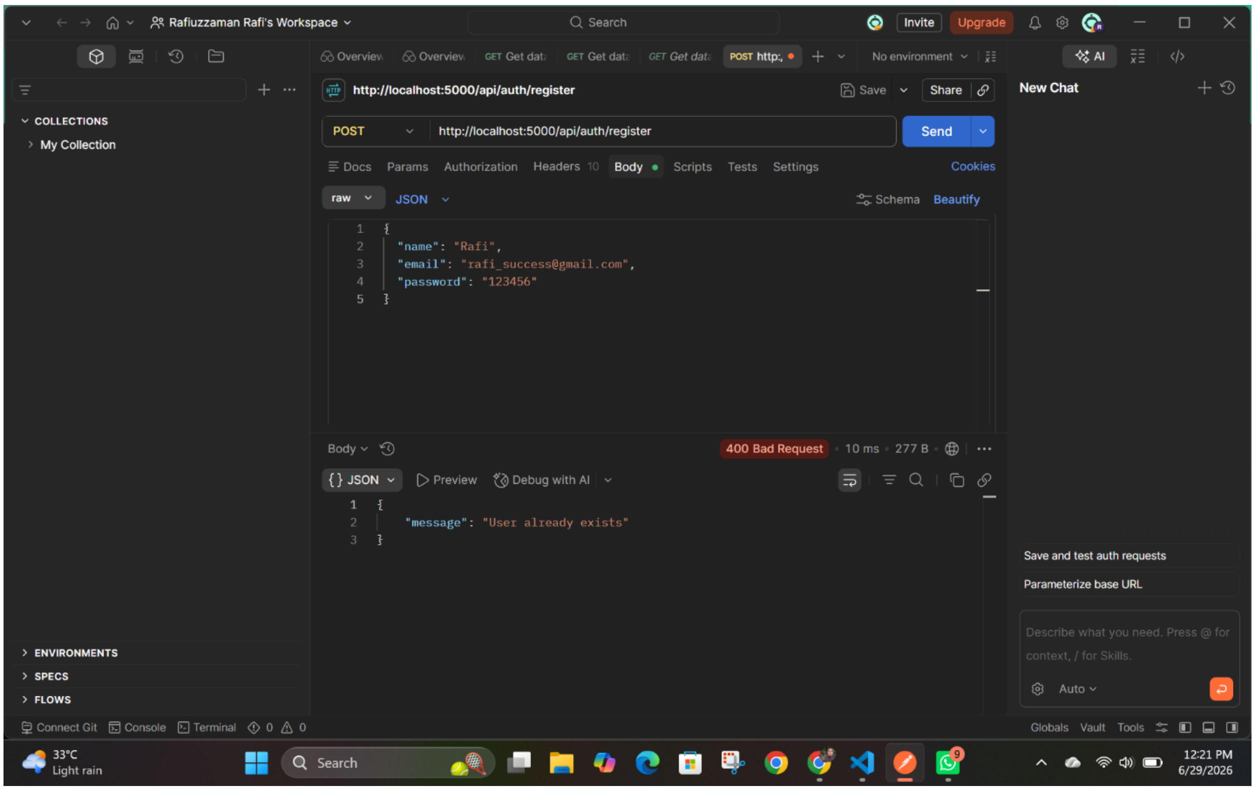
MongoDB connection is configured using Mongoose.



7. Error Handling

The backend implements error handling for:

- Invalid inputs
- Authentication failures
- Duplicate email registration
- Invalid JWT tokens
- Missing files
- Database connection failures
- Internal server errors

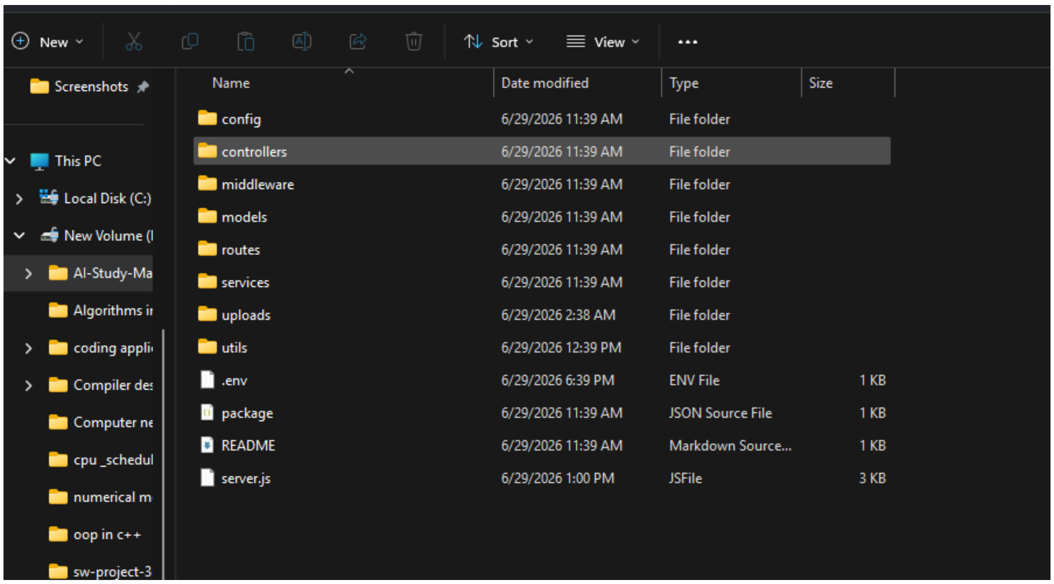


Required Screenshots

Insert the following screenshots:

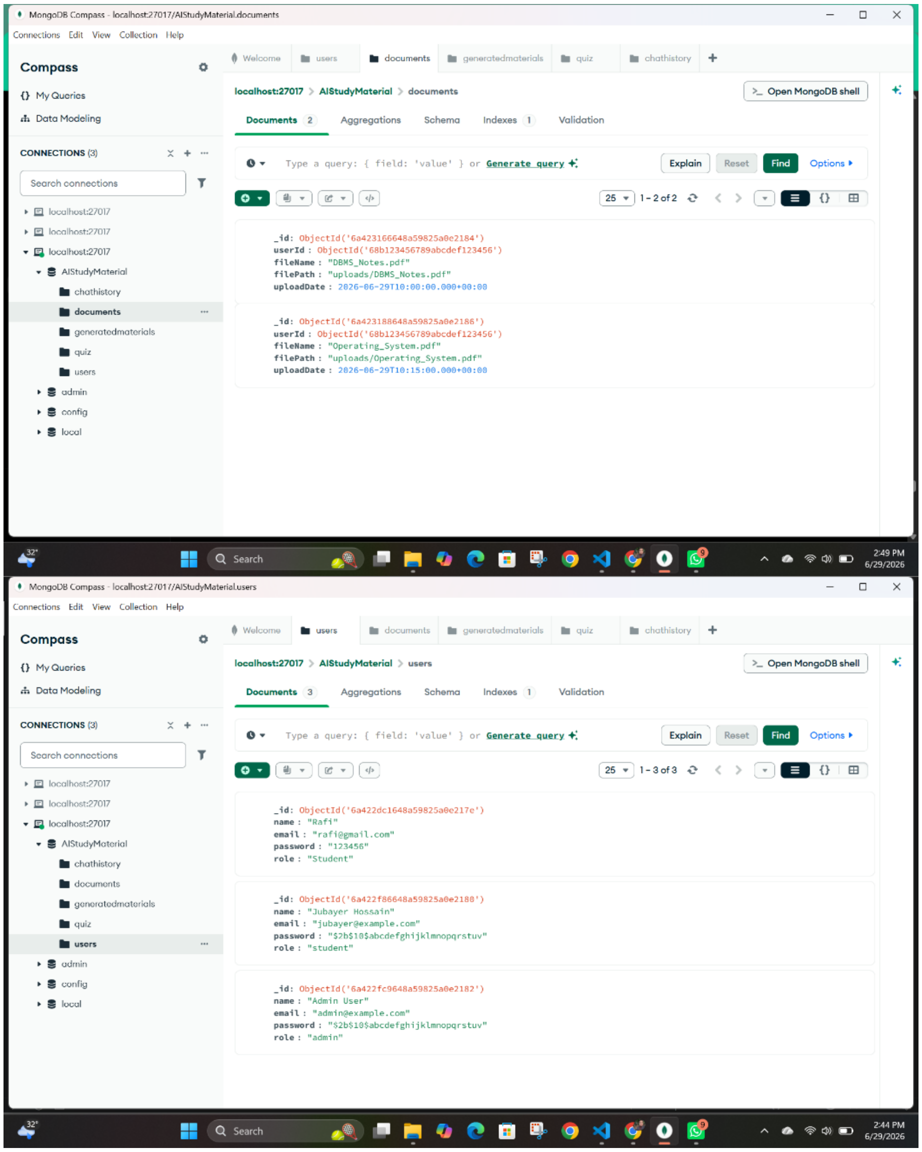
Screenshot 1

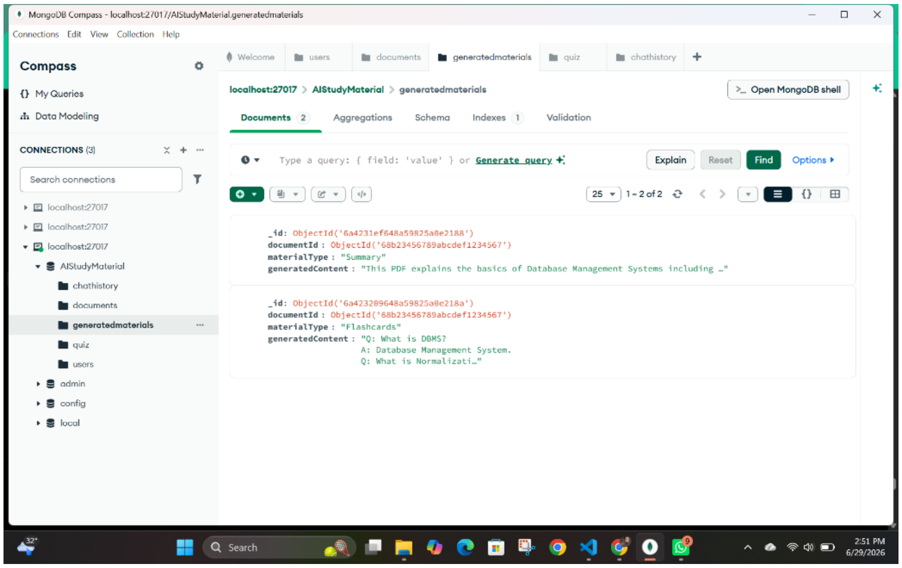
Backend Folder Structure

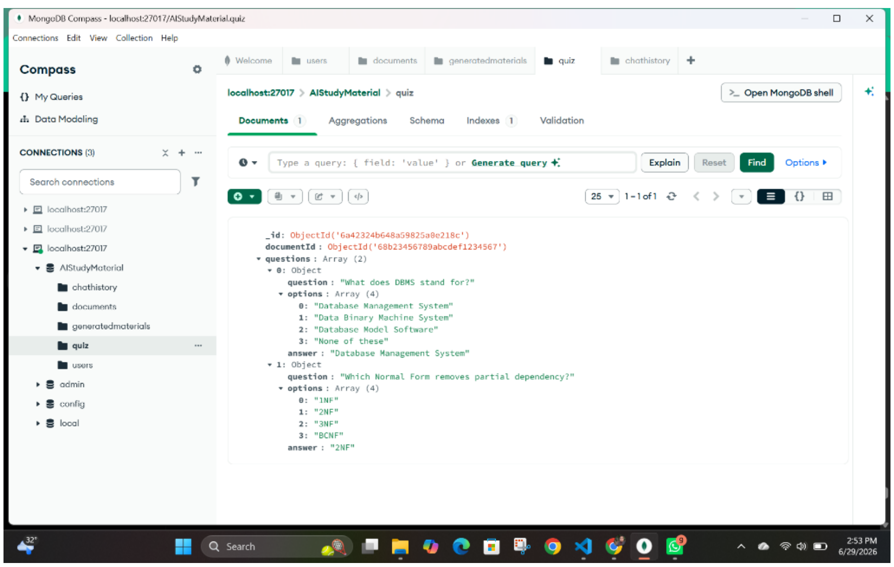


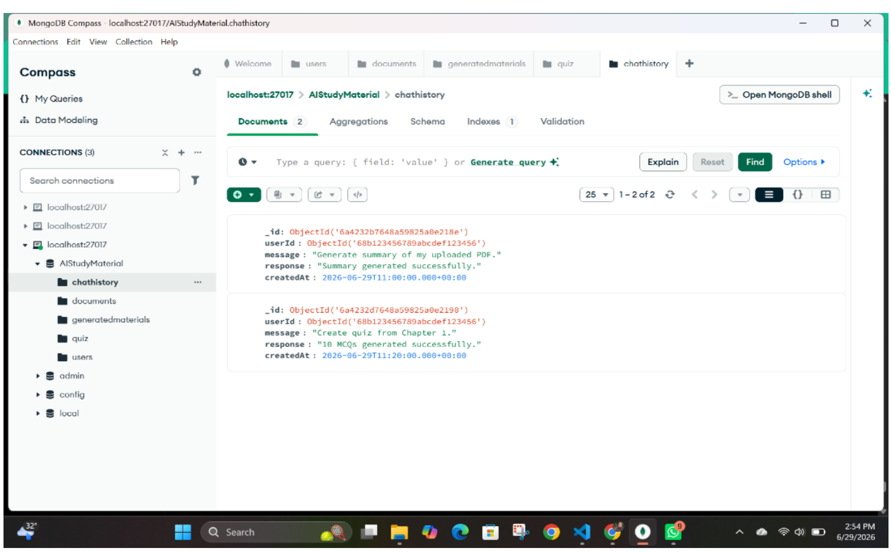
Screenshot 2

MongoDB Collections



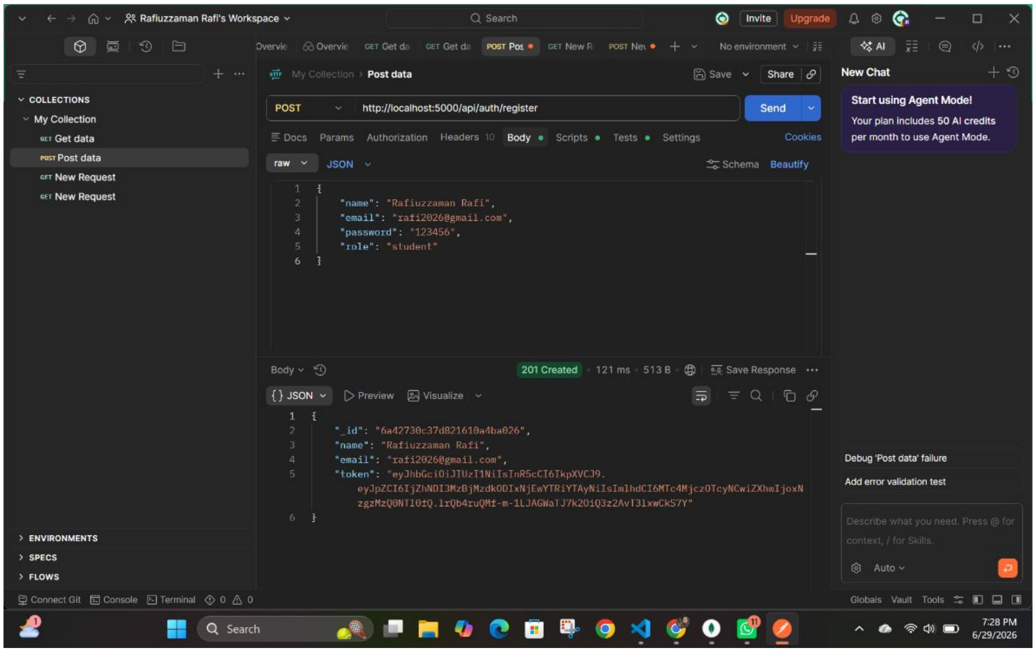

MongoDB Compass - localhost:27017/AIStudyMaterial.generatedmaterials
localhost:27017 > AIStudyMaterial > generatedmaterials
Documents 2
_id: ObjectId('6a4231ef648a59825a0e2188')
documentId : ObjectId('68b23456789abcdef1234567')
materialType : "Summary"
generatedContent : "This PDF explains the basics of Database Management Systems including …"
_id: ObjectId('6a423209648a59825a0e218a')
documentId : ObjectId('68b23456789abcdef1234567')
materialType : "Flashcards"
generatedContent : "Q: What is DBMS?
A: Database Management System.
Q: What is Normalizati…"


MongoDB Compass - localhost:27017/AIStudyMaterial.quiz
localhost:27017 > AIStudyMaterial > quiz
Documents 1
_id: ObjectId('6a42324b648a59825a0e218c')
documentId : ObjectId('68b23456789abcdef1234567')
questions : Array (2)
0: Object
question : "What does DBMS stand for?"
options : Array (4)
0: "Database Management System"
1: "Data Binary Machine System"
2: "Database Model Software"
3: "None of these"
answer : "Database Management System"
1: Object
question : "Which Normal Form removes partial dependency?"
options : Array (4)
0: "1NF"
1: "2NF"
2: "3NF"
3: "BCNF"
answer : "2NF"


MongoDB Compass - localhost:27017/AIStudyMaterial.chathistory
localhost:27017 > AIStudyMaterial > chathistory
Documents 2
_id: ObjectId('6a4232b7648a59825a0e218e')
userId : ObjectId('68b123456789abcdef123456')
message : "Generate summary of my uploaded PDF."
response : "Summary generated successfully."
createdAt : 2026-06-29T11:00:00.000+00:00
_id: ObjectId('6a4232d7648a59825a0e2190')
userId : ObjectId('68b123456789abcdef123456')
message : "Create quiz from Chapter 1."
response : "10 MCQs generated successfully."
createdAt : 2026-06-29T11:20:00.000+00:00

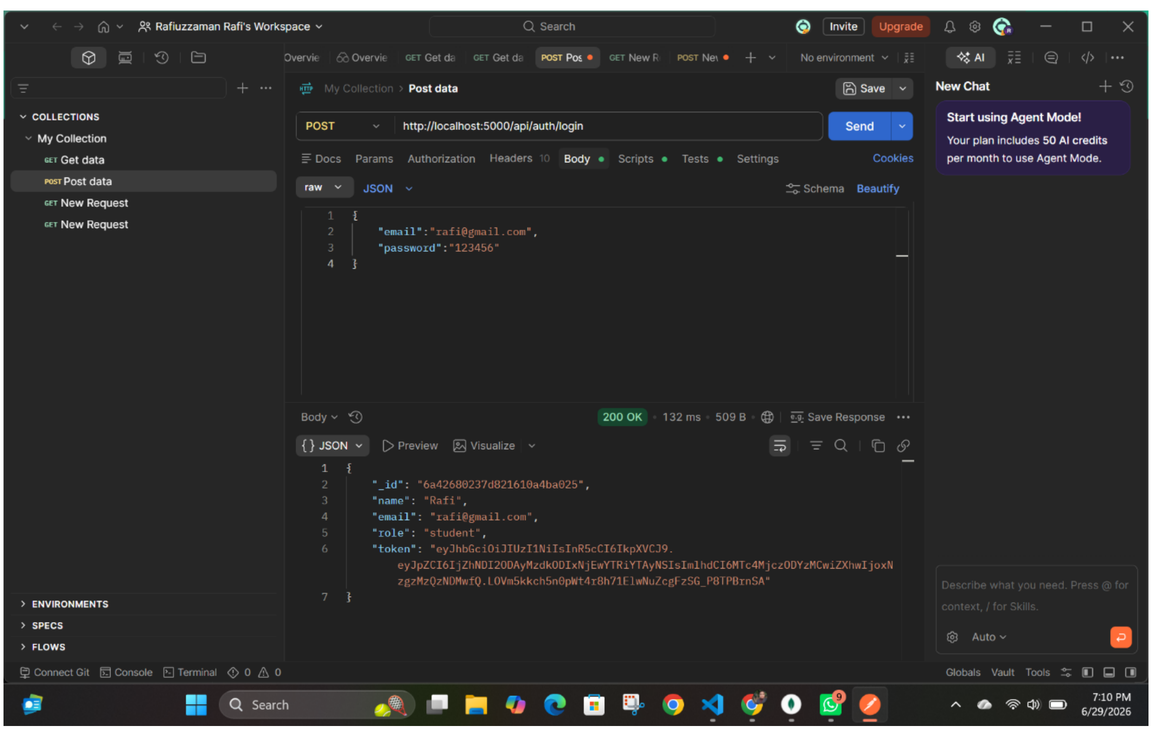
Screenshot 3

User Registration API Test



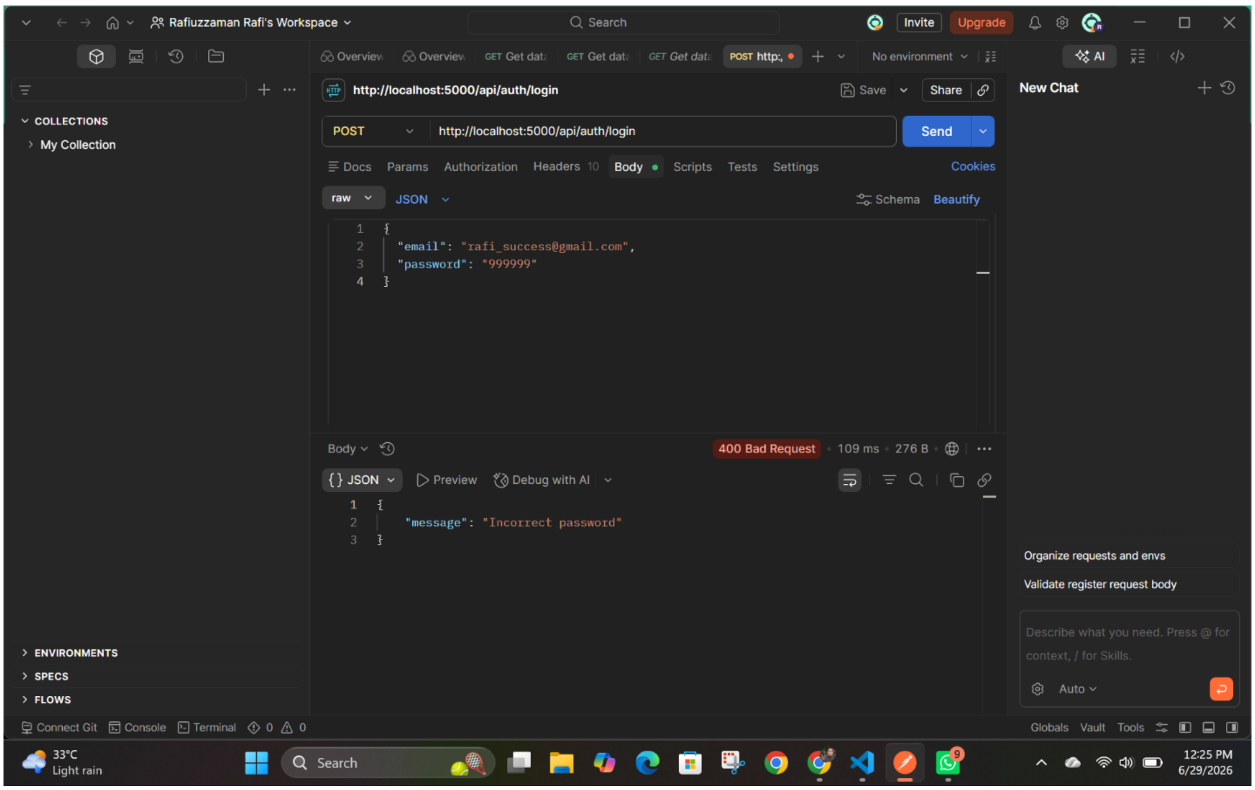
Screenshot 4

User Login API Test



Screenshot 5

Protected API Access Test



Screenshot 6

GitHub Repository